Developmental Neuroanatomy & Congenital Disorders

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Learning Objectives

1. Describe the embryologic development of the central nervous system, including neurulation, neural tube formation, and differentiation of the brain and spinal cord, and relate these processes to mature neuroanatomy. (CLO 1)
2. Describe the embryologic development of the pharyngeal apparatus and relate its derivatives to clinically relevant head and neck structures. (CLO 1)
3. Correlate abnormalities in neural and craniofacial development with common congenital disorders involving the central nervous system and head and neck. (CLO 3)

Pre-Session Assignment

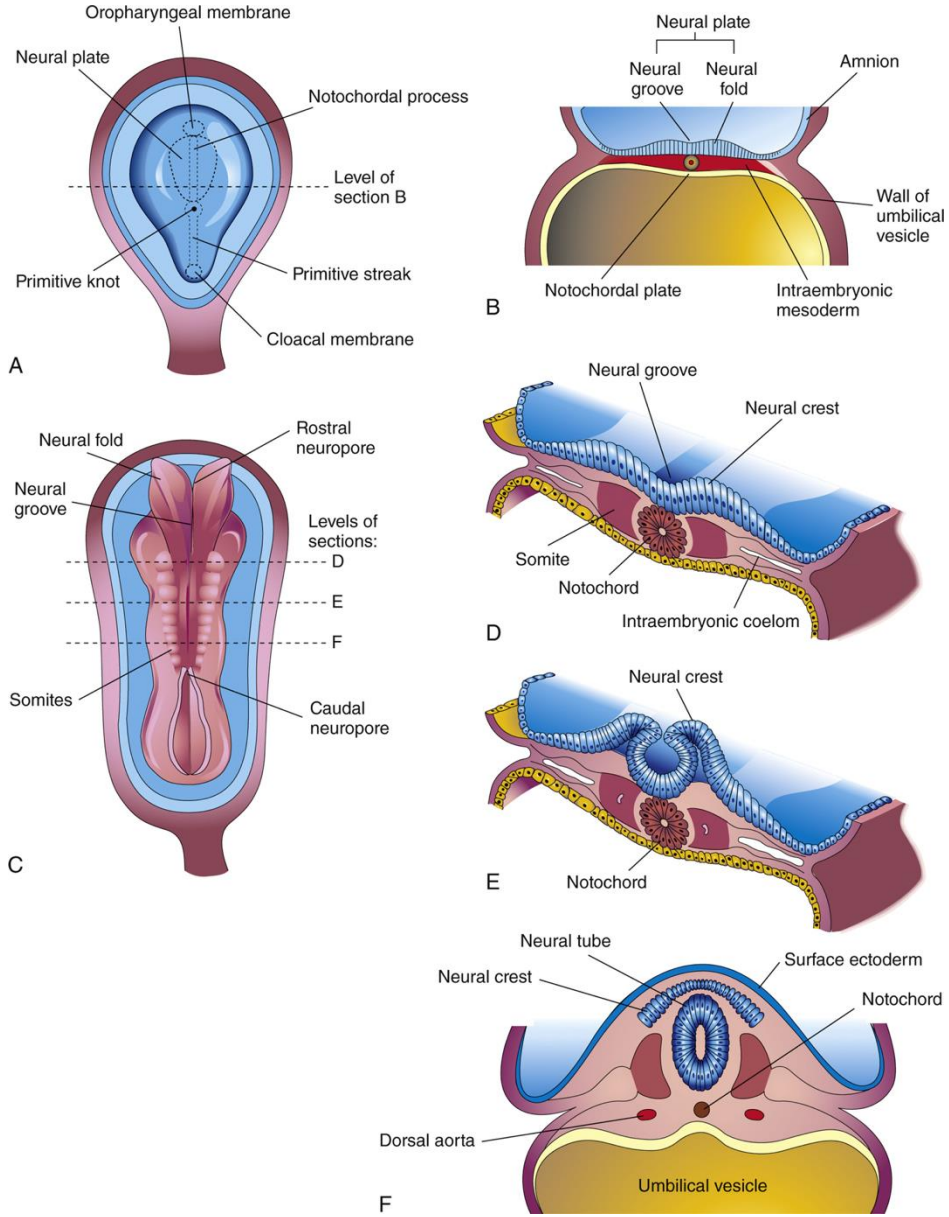
- Preview the PowerPoint slides. You are not expected to memorize details before class.
- Watch: B&B: Neuroembryology (29 min)

Development of the Neural Tube

- Begins in week 3 → neural plate & groove
- Notochord induces ectoderm → neural plate
- Signals: TGF-β, Wnts, SHH, BMPs
- Neural folds fuse → neural tube (CNS)
- Neural crest → PNS & ANS structures

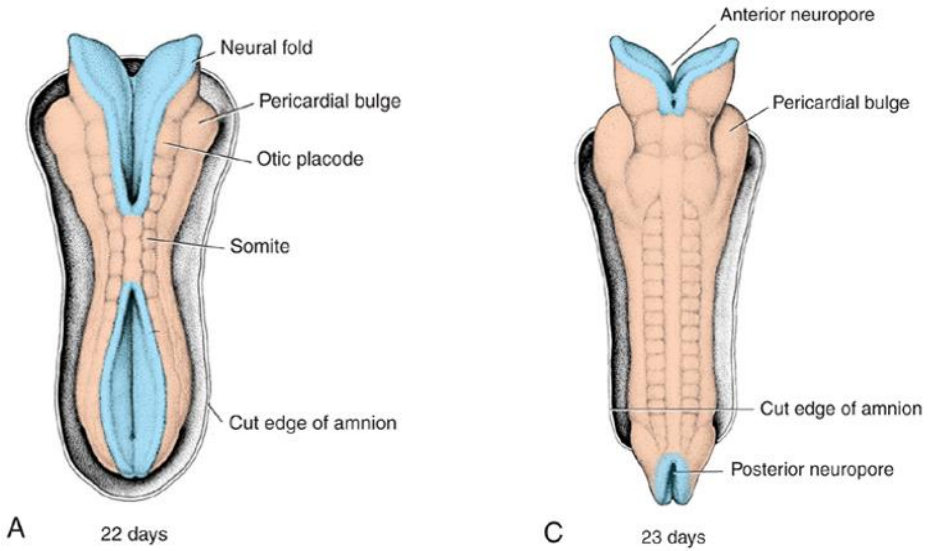
Sequence (see image)

- Neural plate
- Neural groove + folds
- Fusion → neural tube + neural crest



Neural plate and folding leading to neural tube formation. (A) Neural plate (day 17). (B) Early neural groove and folds. (C) Fusion begins (day 22), still open at both ends. (D–F) Transverse sections show closure, tube separation from ectoderm, and neural crest formation. Fig 16.1 BWAB, 11e.

Neurulation (Week 4)



Neural tube first closes in the future cervical region.
The neural folds close in zipper-like fashion in both directions.

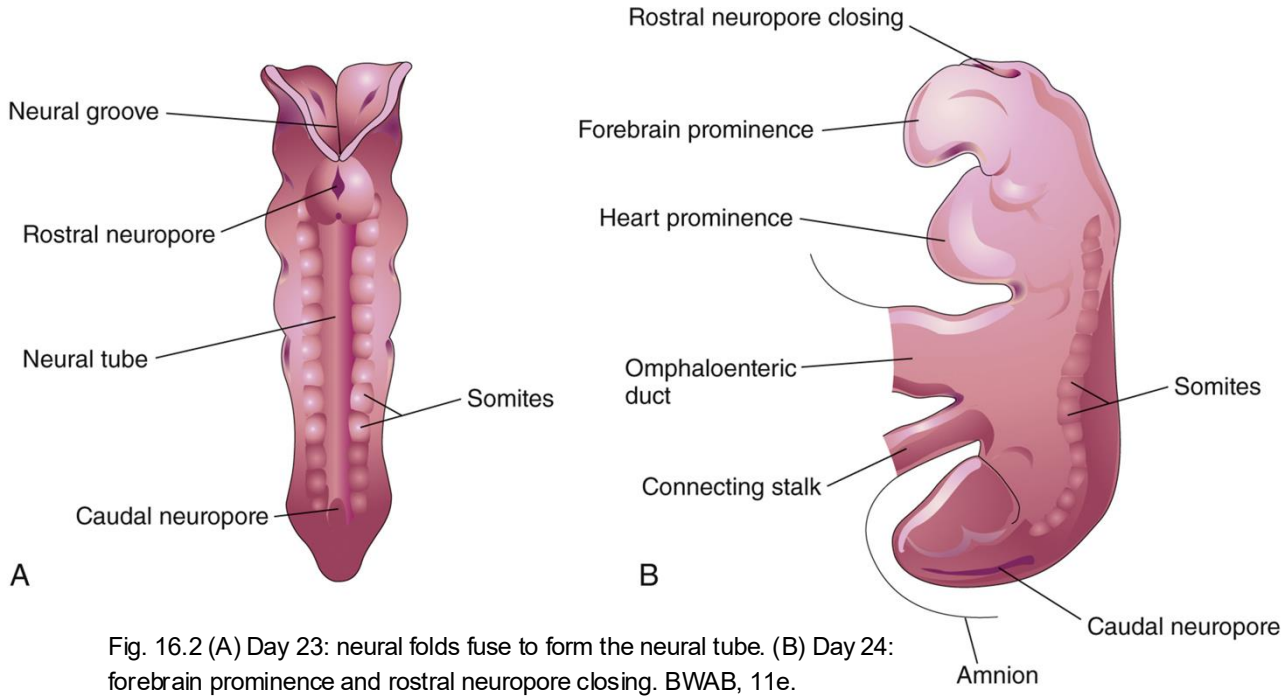


Fig. 16.2 (A) Day 23: neural folds fuse to form the neural tube. (B) Day 24: forebrain prominence and rostral neuropore closing. BWAB, 11e.

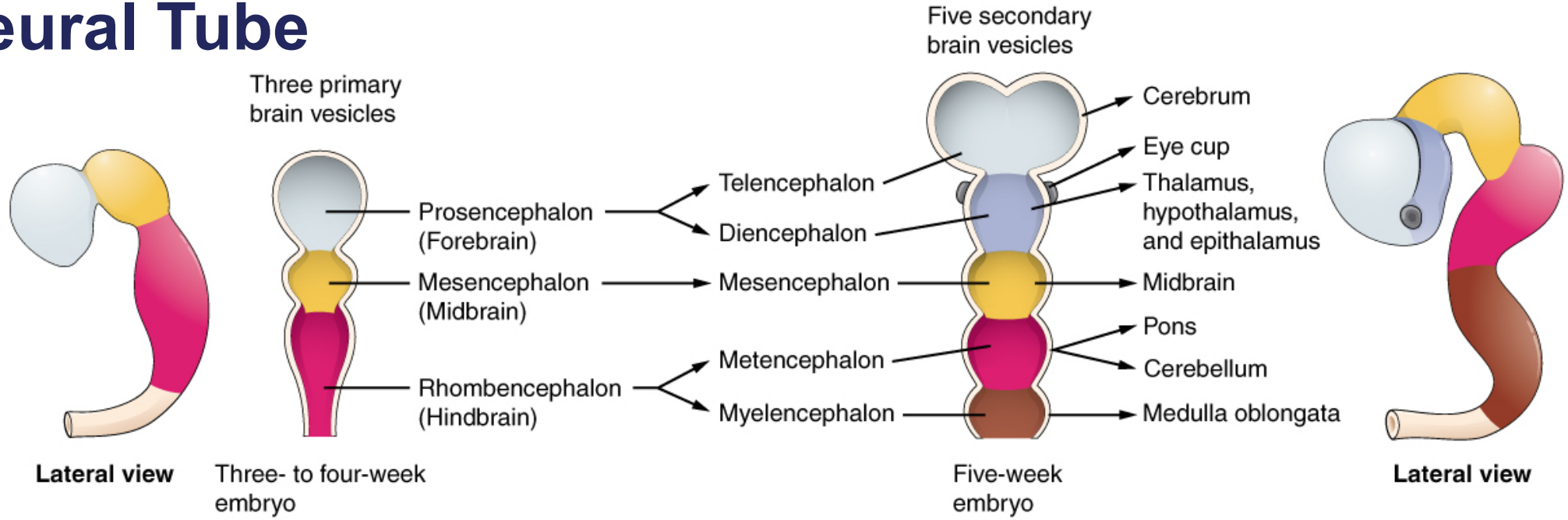
- Neural folds fuse → neural tube
- Neural canal is open at both ends

- Rostral neuropore closes ~day 25
- Caudal neuropore closes ~day 27

Neural tube → Brain and Spinal Cord

Neural Canal → Ventricular System

Neural Tube

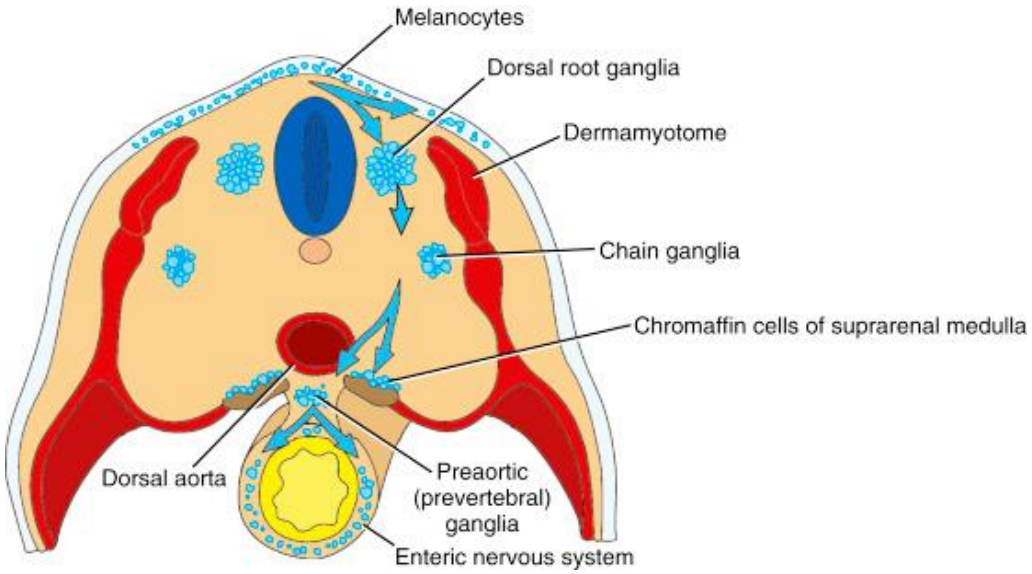


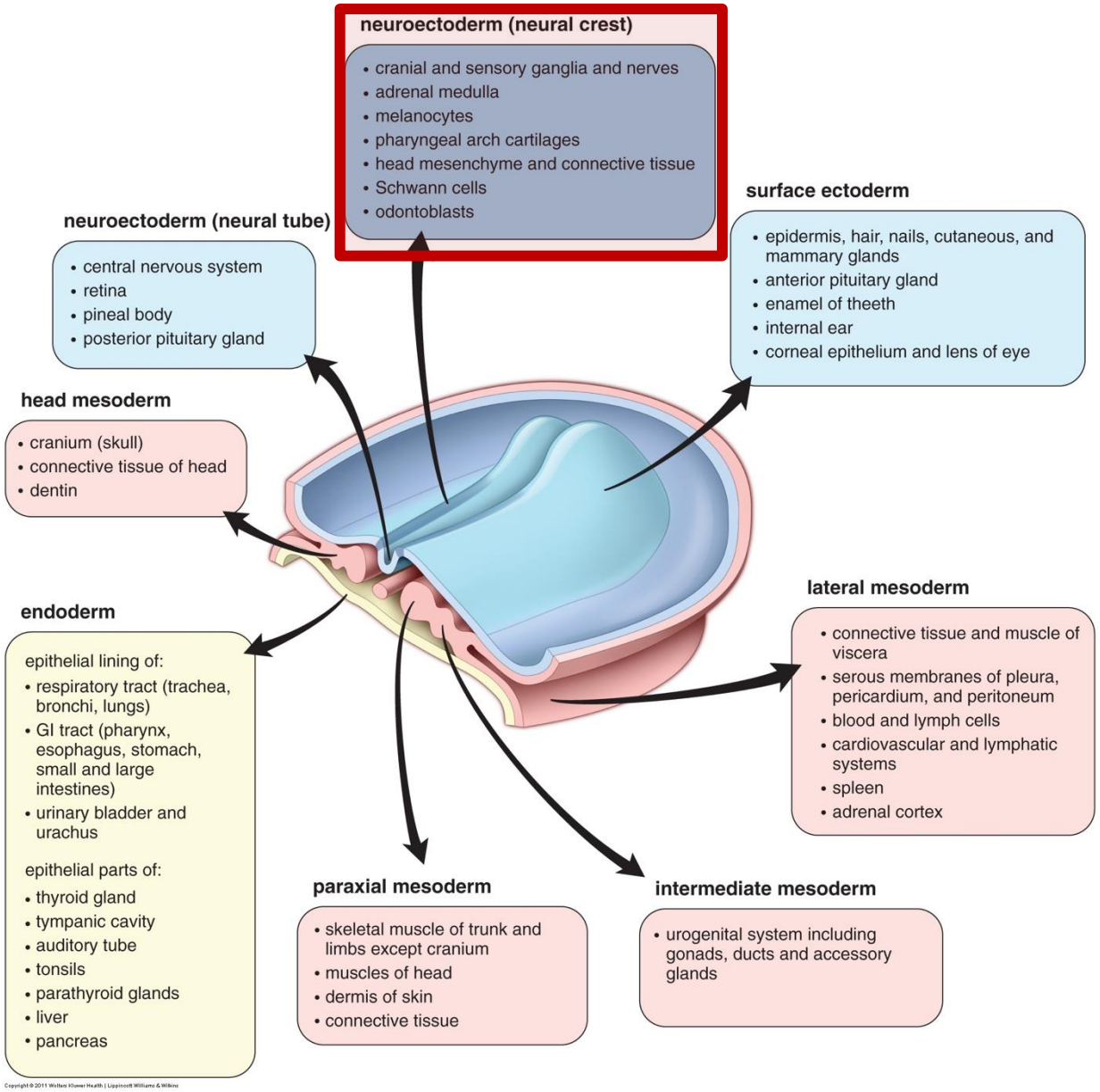
Primary vesicle	Secondary vesicle	Neural derivatives	Cavity
Prosencephalon (forebrain)	Telencephalon	Cerebral hemispheres	Lateral ventricles
	Diencephalon	Thalamus, hypothalamus, epithalamus, subthalamus, retina	Third ventricle
Mesencephalon (midbrain)	Mesencephalon	Midbrain	Cerebral aqueduct
Rhombencephalon (hindbrain)	Metencephalon	Pons, cerebellum	Part of fourth ventricle
	Myelencephalon	Medulla	Part of fourth ventricle, part of central canal

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Neural Crest

- Defined at the neural plate border during neurulation
- After tube closure, cells undergo EMT → migrate widely





EMT = epithelial-to-mesenchymal transition

Failure of Neural Tube Closure



Carlson: Human Embryology and Developmental Biology, 5e
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Anencephaly
Failure of closure of anterior neuropore



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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Neural Tube Defect
Failure of closure of posterior neuropore

Images: Carlson Fig 8-4, Larsen Fig 4-11

Spina Bifida

Spina Bifida Occulta

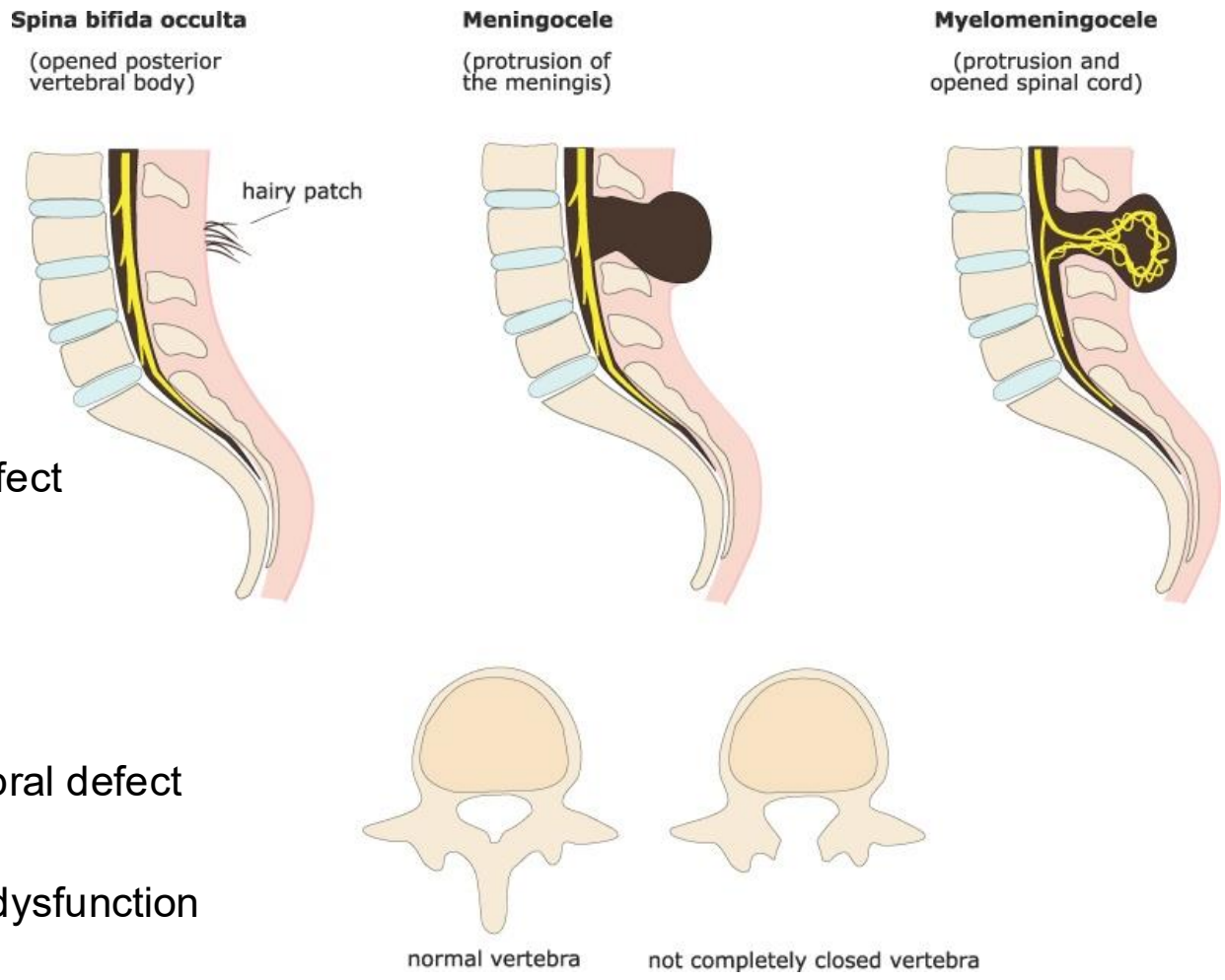
- Failure of fusion of posterior vertebral arches
- Spinal cord and meninges remain intact
- Usually asymptomatic; may have overlying tuft of hair
- Typically lumbosacral; often discovered incidentally

Meningocele

- Herniation of meninges and CSF through vertebral defect
- No spinal cord tissue in the sac
- Neurological deficits usually absent

Myelomeningocele

- Herniation of meninges and spinal cord through vertebral defect
- Most commonly lumbosacral
- Lower limb motor/sensory deficits and bowel/bladder dysfunction
- Risk of infection if neural tissue is exposed



Occulta: Bony defect only | Meningocele: Meninges + CSF | Myelomeningocele: Meninges + spinal cord



Cervical Meningocele



Myelomeningocele

Internet Scientific Publication
Moore. Before We Are Born

Neural Tube Defects (NTDs): Risk Factors & Prevention

Neural Tube Closure: Normally completed by day 28 of embryonic development

Folate deficiency: Major risk factor during early pregnancy

Folic acid supplementation: Before conception and during early pregnancy reduces NTD risk

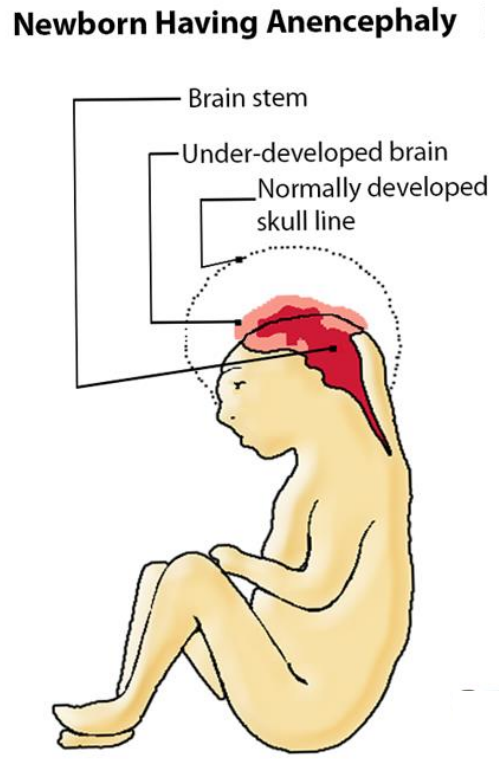
Antiseizure medications: Valproic acid and carbamazepine increase NTD risk

Genetic susceptibility: increased concordance in monozygotic twins

Anencephaly & Holoprosencephaly

Anencephaly

- Failure of anterior (rostral) neuropore closure during week 4
- Absence of most of the cerebral hemispheres and calvarium
- Severe, typically lethal neural tube defect
- Associated with folate deficiency



Holoprosencephaly

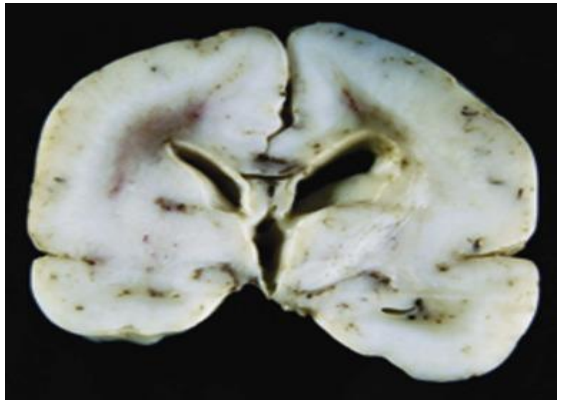
- Failure of the prosencephalon to divide into cerebral hemispheres
- Severe forms associated with midline facial defects (e.g., cyclopia)
- Associated with SHH signaling abnormalities and trisomy 13

Anencephaly = failure of neural tube closure | Holoprosencephaly = failure of forebrain division

Other Disorders of Brain Development

Lissencephaly

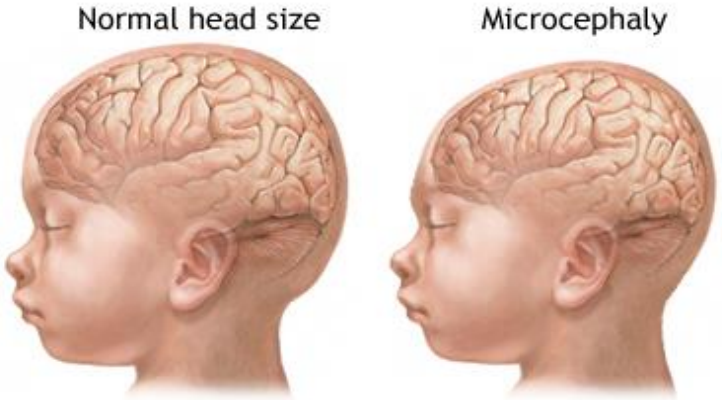
- Abnormal neuronal migration → reduced or absent cerebral gyri (smooth brain)
- Associated with developmental delay, seizures, and intellectual disability
- Classic genetic associations: LIS1 (PAFAH1B1) and DCX



Robbins and Cotran Pathology

Microcephaly

- Abnormally small head circumference due to impaired brain growth
- Associated with genetic abnormalities and congenital infections (e.g., Zika virus)
- May cause developmental delay and intellectual disability



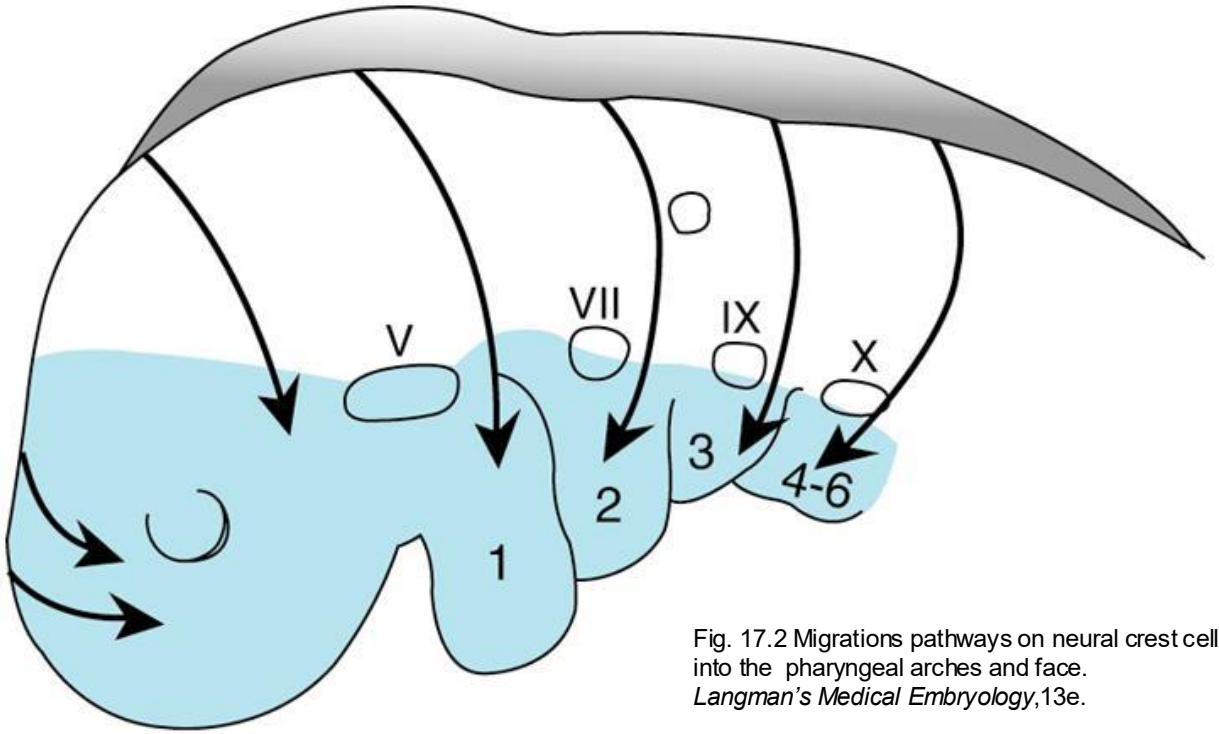
Adam

Lissencephaly = abnormal neuronal migration

Microcephaly = impaired brain growth

Pharyngeal Apparatus

Neural Crest → Pharyngeal Arches



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Fig. 17.2 Migrations pathways on neural crest cells into the pharyngeal arches and face. *Langman's Medical Embryology*,13e.

- Neural crest cells migrate from neural tube
- Populate pharyngeal arches, which form head & neck structures
- Give rise to skeletal, vascular, and connective tissue elements

Pharyngeal Apparatus Overview

- Components = arches + pouches + clefts
- Timing: forms during weeks 4–7
- Function: gives rise to most head & neck structures
- Clinical note: errors → congenital anomalies

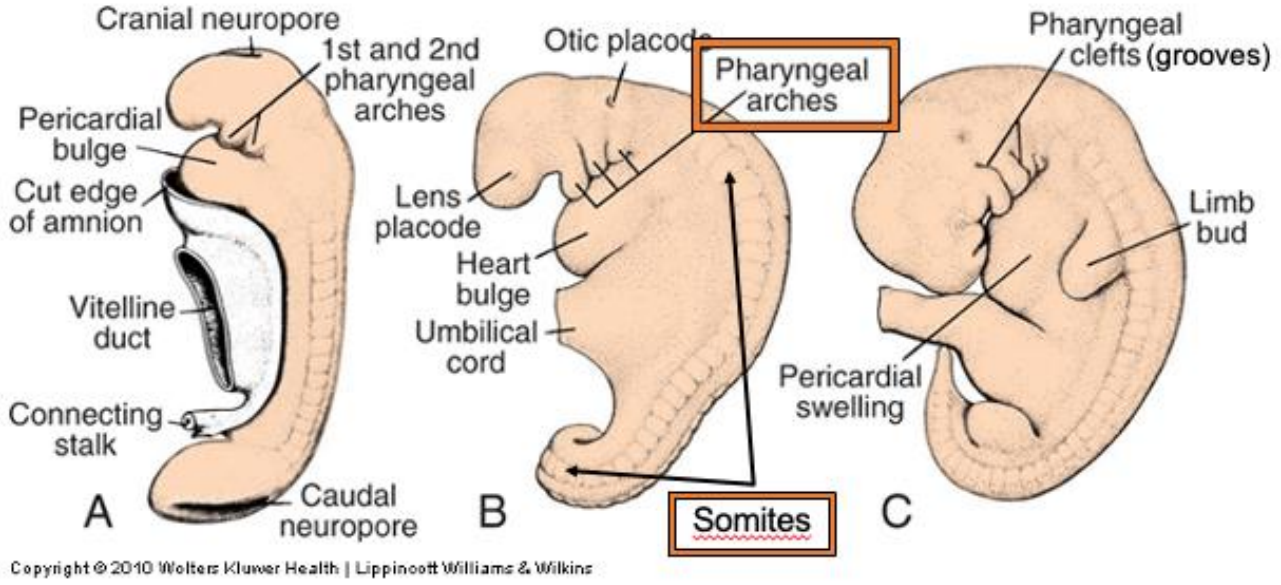


Fig. 17.3 Development of the Pharyngeal arches. A. 25 days. B. 28 Days. C. 5 weeks
Langman's Medical Embryology,13e.

Arch 1 (Mandibular Arch)

- **Skeletal Derivatives:** mandible, malleus, incus
- **Muscles:** mastication, mylohyoid, anterior digastric, tensors (tensor tympani, tensor veli palatini)
- **Nerve:** CN V_3 (mandibular nerve)
- Clinical: Treacher Collins (first arch defect; mandibular hypoplasia, hearing loss)

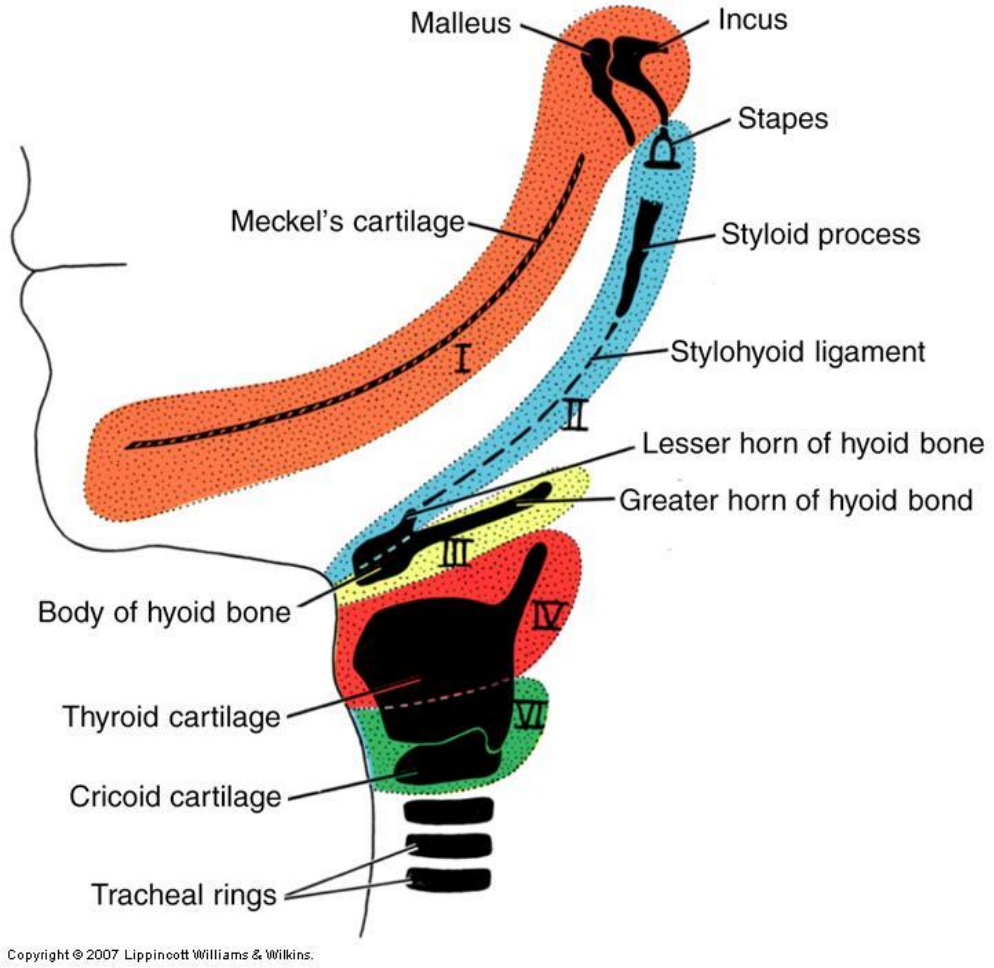


Fig. 17.9 Definitive structures formed by the cartilaginous components of the various pharyngeal arches. *Langman's Medical Embryology*,13e.

Arch 2 (Hyoid Arch)

- **Skeletal Derivatives:** stapes, styloid, lesser hyoid horn
- **Muscles:** Facial expression, stapedius, posterior digastric, stylohyoid
- **Nerve:** CN VII
- **Pathology:** Conductive hearing loss (stapes fixation)

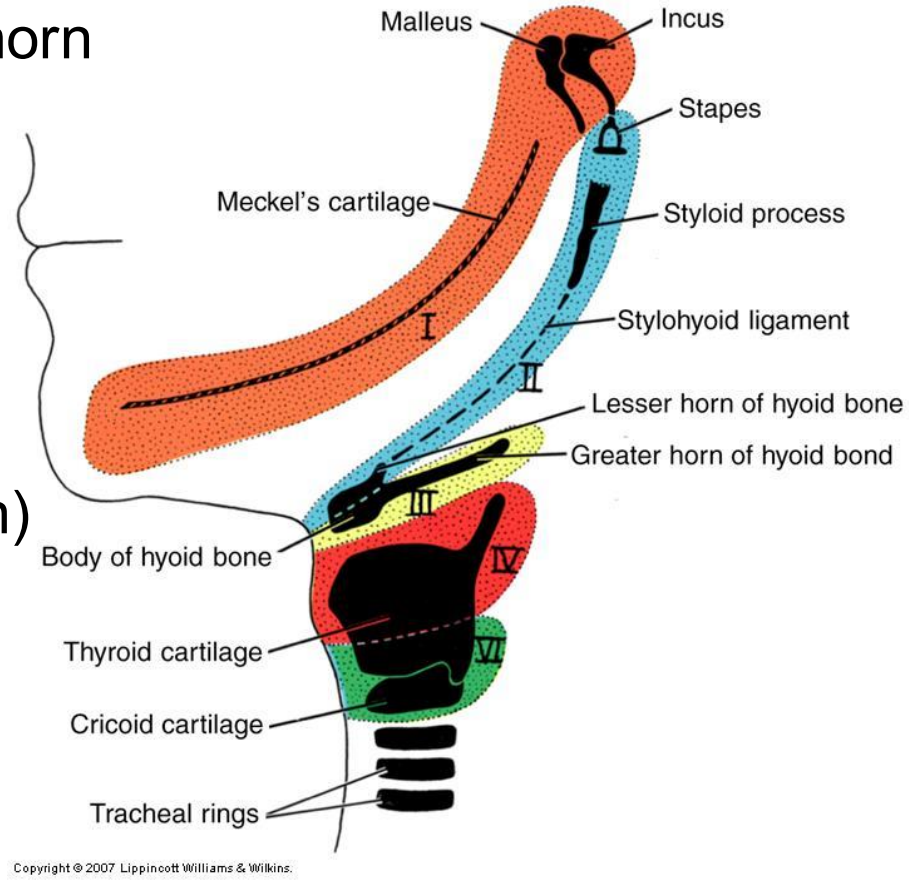


Fig. 17.9 Definitive structures formed by the cartilaginous components of the various pharyngeal arches. *Langman's Medical Embryology*,13e.

Arch 3 (Glossopharyngeal Arch)

- **Skeletal Derivative:** Greater hyoid horn
- **Muscle:** Stylopharyngeus
- **Nerve:** CN IX
- **Pathology:** Loss of gag reflex (afferent)

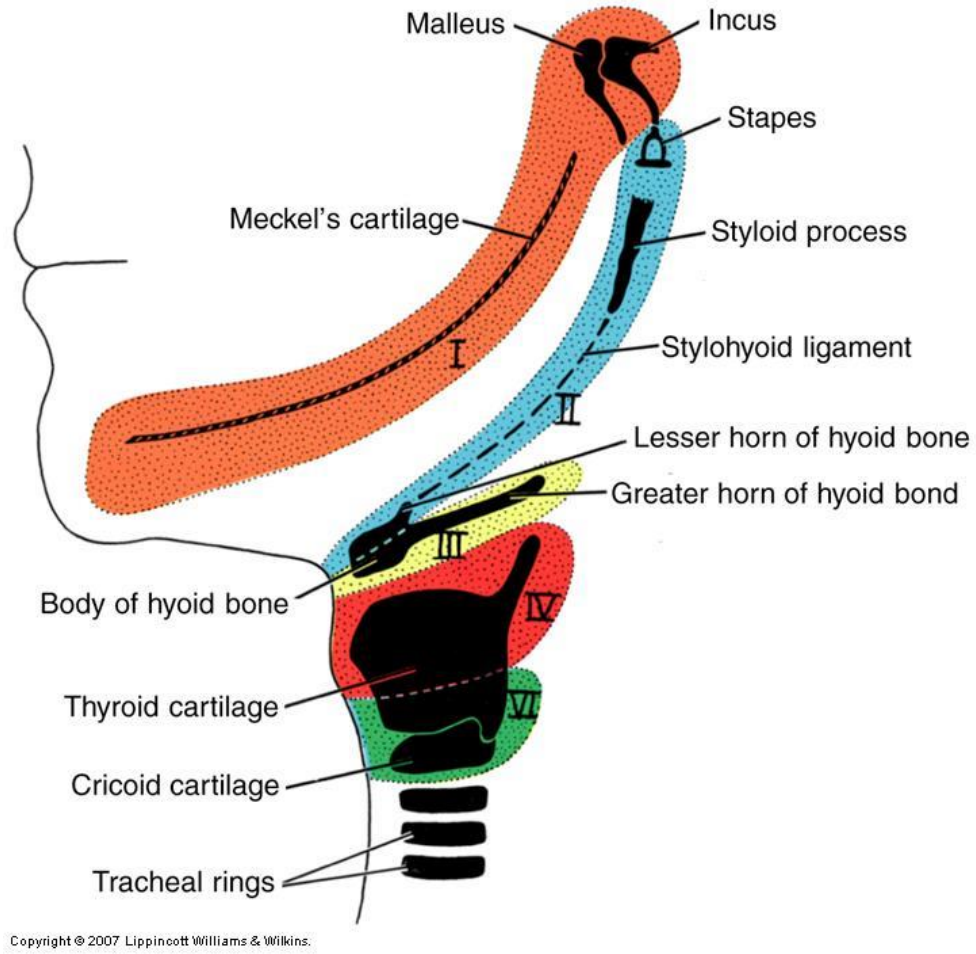


Fig. 17.9 Definitive structures formed by the cartilaginous components of the various pharyngeal arches. *Langman's Medical Embryology*,13e.

Arches 4 & 6 (Laryngeal Arches)

- **Skeletal Derivatives:** Laryngeal framework
- **Muscles:** muscles of larynx, most muscles of pharynx and soft palate
- **Nerve:** CN X
- **Pathology:** dysphagia, pharyngeal weakness, hoarseness

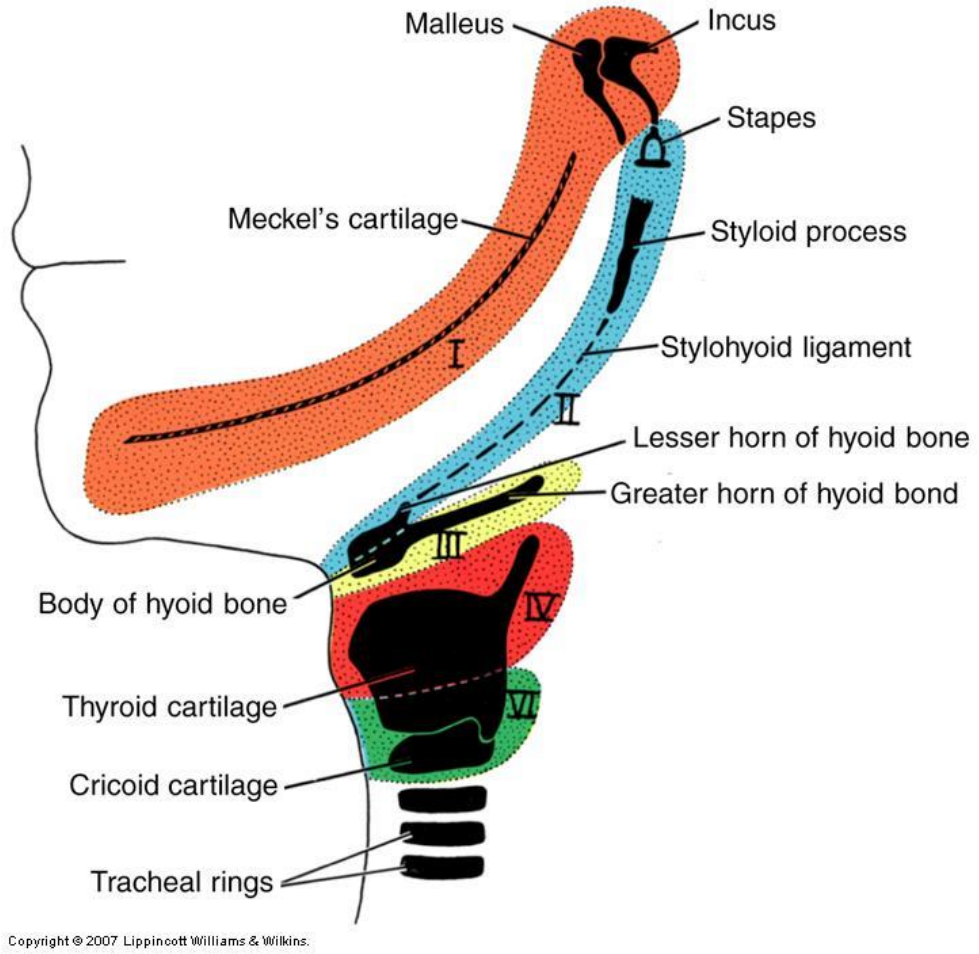
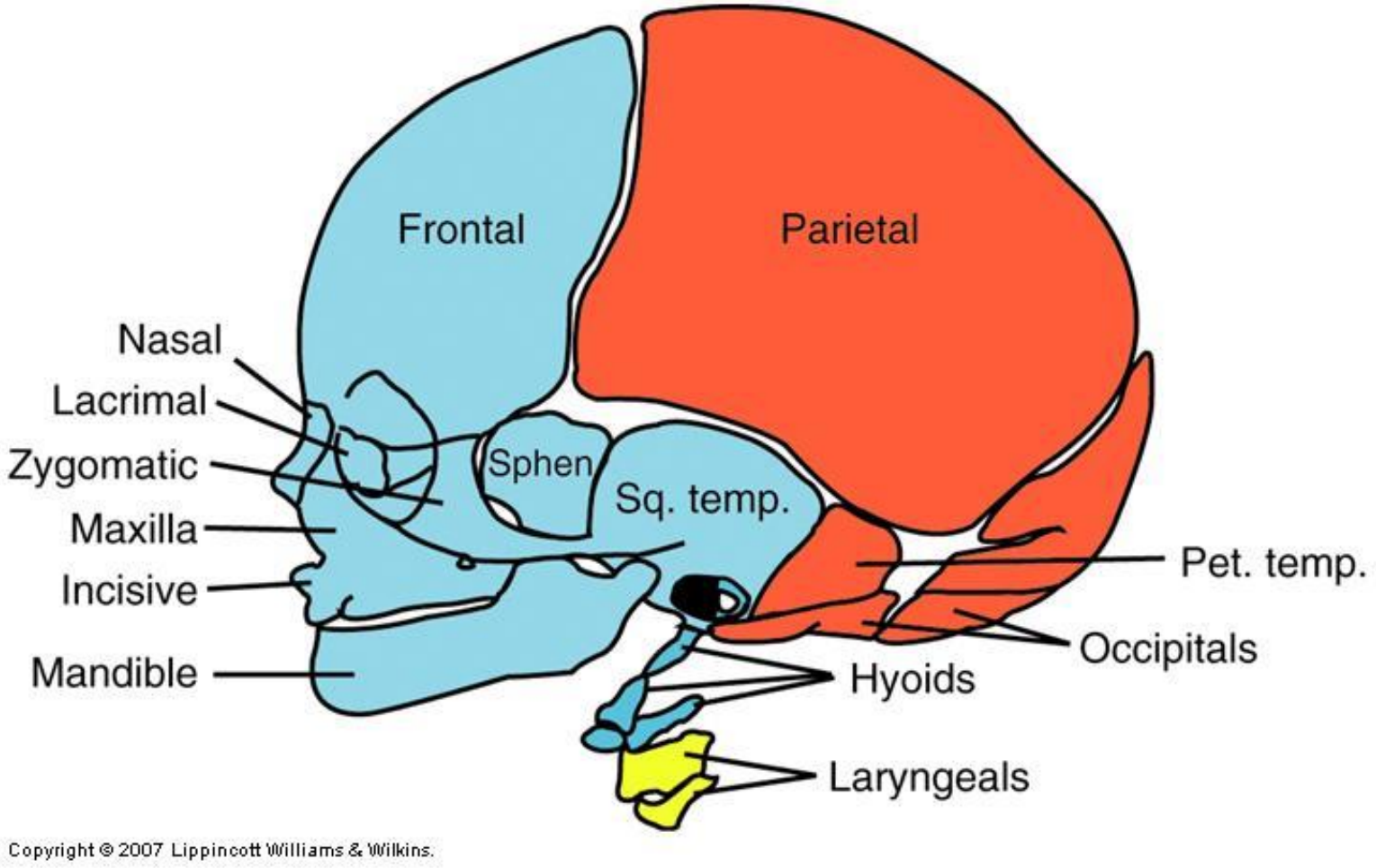


Fig. 17.9 Definitive structures formed by the cartilaginous components of the various pharyngeal arches. *Langman's Medical Embryology*,13e.

Skeletal Structures of the Head and Face

Blue structures from neural crest

Red structures from occipital somites



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Fig. 17.1 Skeletal structures of the head and face. *Langman's Medical Embryology*,13e.

Agnathia

- Cause: failure of neural crest cells to populate 1st arch
- Severe 1st arch malformation (extreme hypoplasia)
- Mandible fails to form (absent lower jaw)
- Rare, often incompatible with life



Treacher Collins Syndrome – Clinical Features

(Mandibulofacial Dysostosis; 1st Arch Syndrome)

- **Cause:** Abnormal development and survival of neural crest cells contributing to the first pharyngeal arch
- **Genetics:** Most commonly TCOF1 mutation; usually autosomal dominant

Craniofacial Abnormalities

- Mandibular and zygomatic hypoplasia
- Downward-slanting palpebral fissures
- External ear and ossicular malformations
- Clinical: Conductive hearing loss; variable severity

Failure of first arch neural crest migration
→ craniofacial malformations + conductive hearing loss

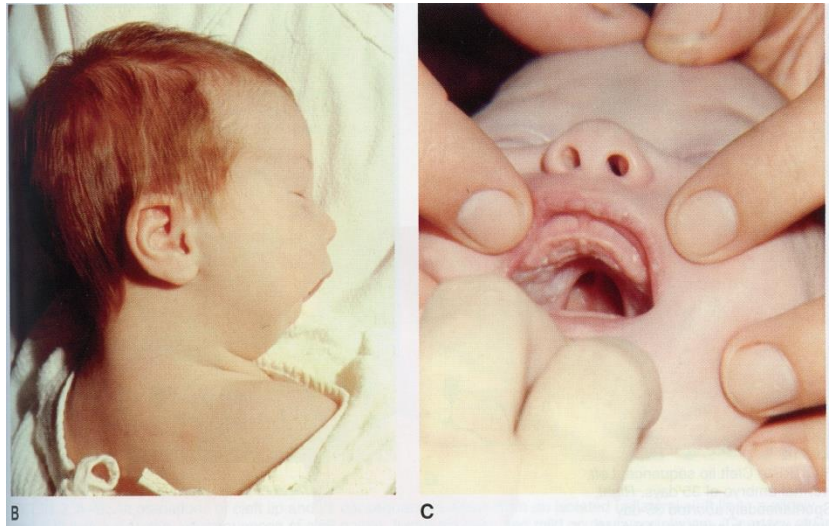


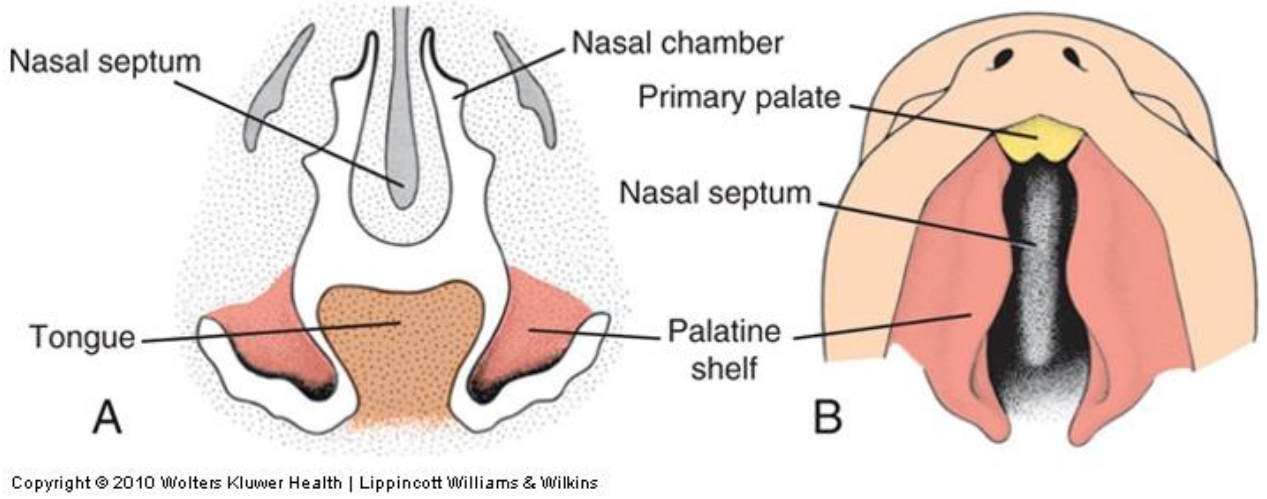
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Pierre Robin Sequence

Sequence

- Initial abnormality: Mandibular hypoplasia (micrognathia)
- Glossoptosis: Posterior displacement of the tongue
- Cleft palate: Tongue interferes with normal palatal shelf elevation and fusion
- Clinical: Upper airway obstruction and feeding difficulties





Micrognathia → Glossoptosis → Cleft Palate

Pharyngeal Apparatus: Arches, Pouches & Clefts

Pharyngeal Arches

- External ectoderm, internal endoderm, and mesenchymal core
- Each arch contains cartilage, muscle, nerve, and artery components

Pharyngeal Pouches

- Endoderm-lined internal out-pocketings
- Form structures including the thymus, parathyroids, and middle ear cavity

Pharyngeal Clefts

- Ectoderm-lined external grooves
- First cleft → external acoustic meatus
- Remaining clefts normally disappear

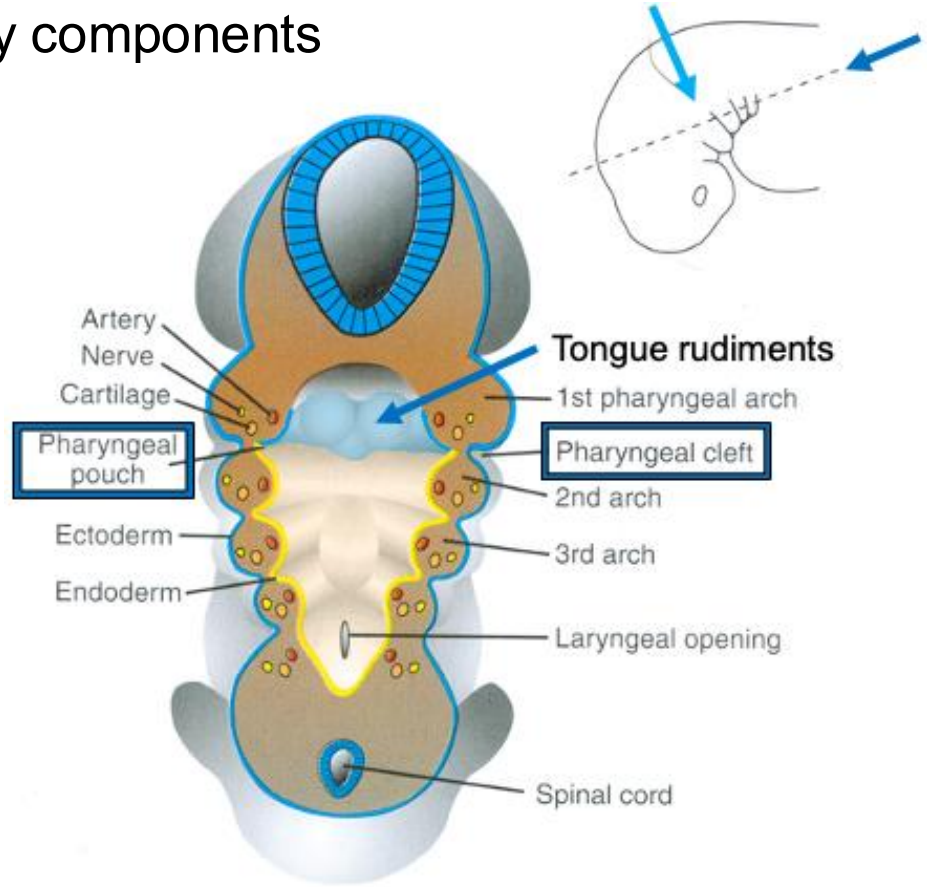


Fig. 17.6 Pharyngeal arches sectioned
Langman's Medical Embryology,13e.
(Modified with Photoshop to show correct first arch germ layer.)

Pouches

- **1st Pouch:** Middle ear, Eustachian tube
- **2nd Pouch:** Palatine tonsils
- **3rd Pouch:** Thymus, inferior parathyroids
- **4th Pouch:** Superior parathyroids, C cells (via ultimobranchial body)

Clinical

- DiGeorge (3rd/4th pouch) → T-cell deficiency, hypocalcemia

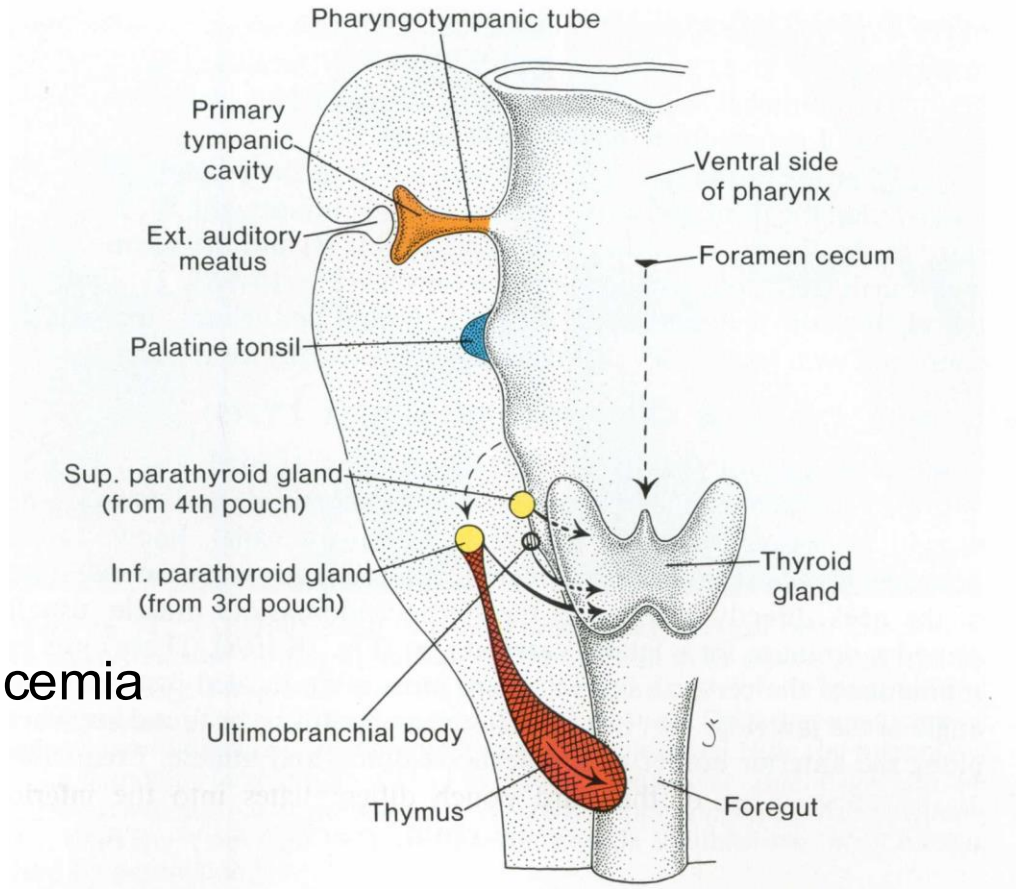


Fig. 17.11 Migration of the thymus, parathyroid glands and the ultimobranchial body. *Langman's Medical Embryology*,13e.

Branchial (Pharyngeal) Clefts

- 1st cleft → external auditory canal
- 2nd–4th clefts → normally obliterate
- Clinical: persistence → branchial cleft cysts / fistula

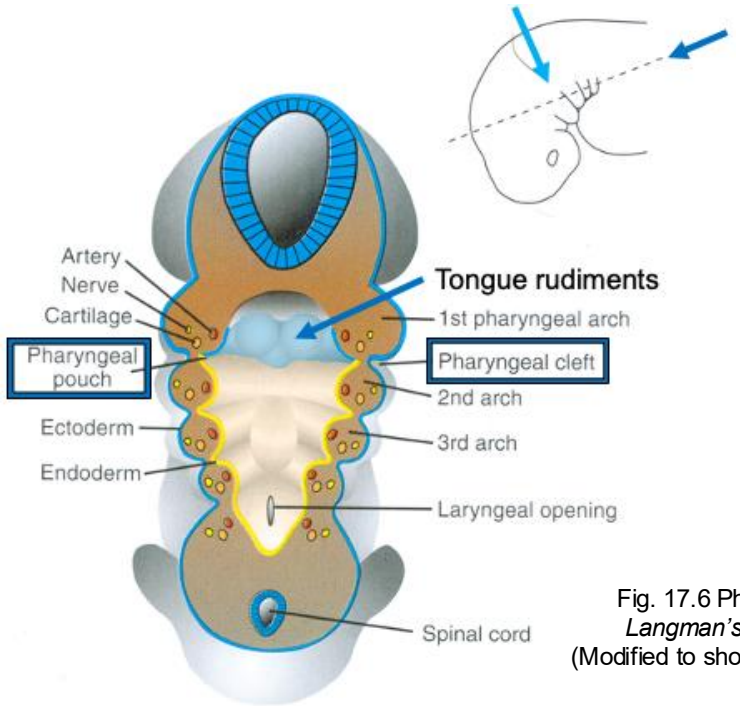


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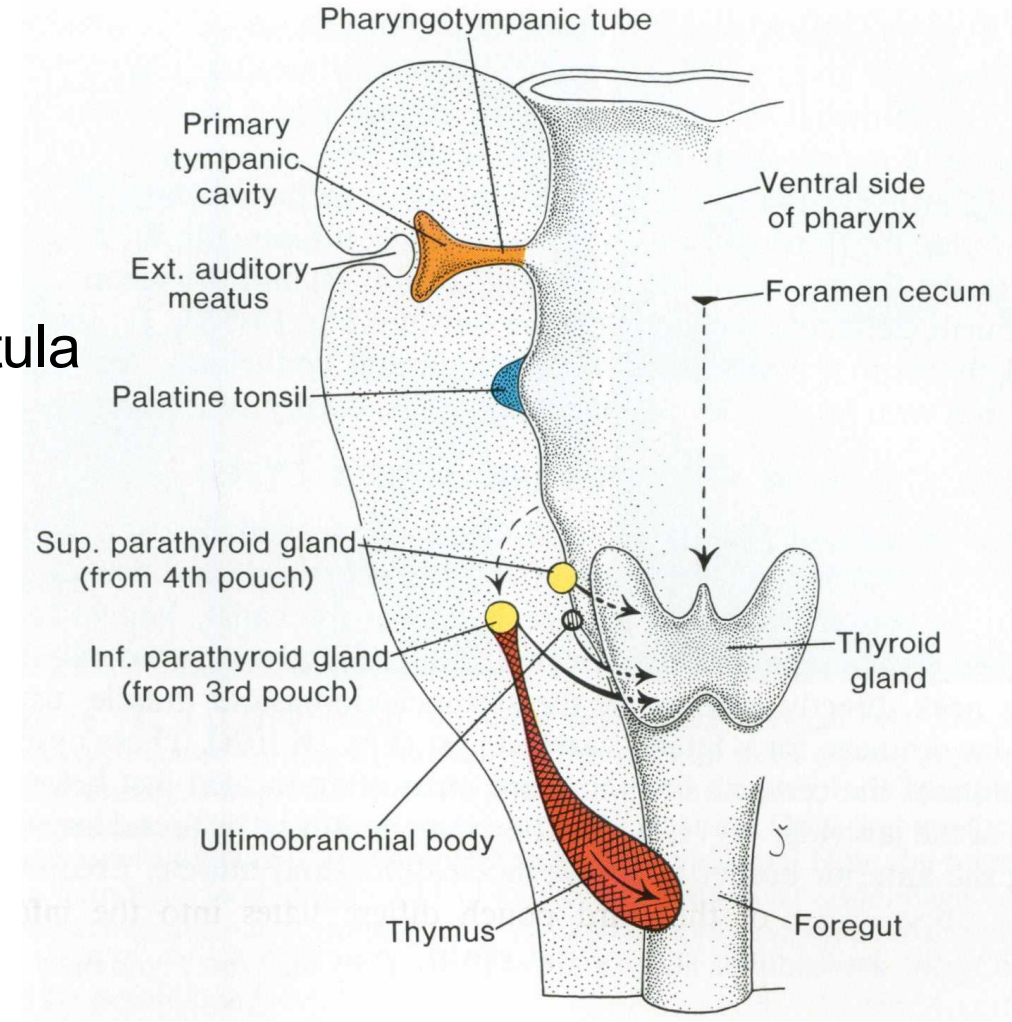
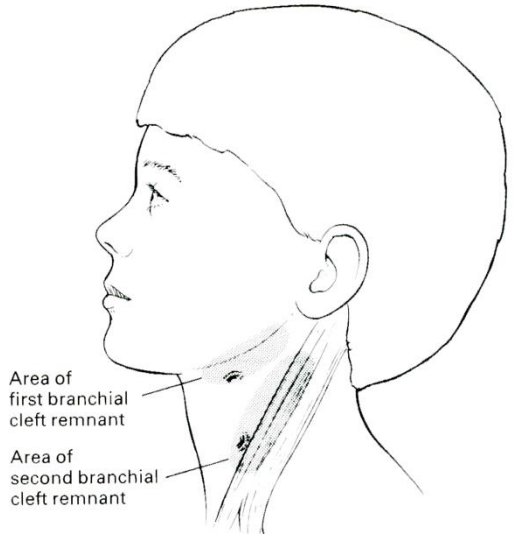


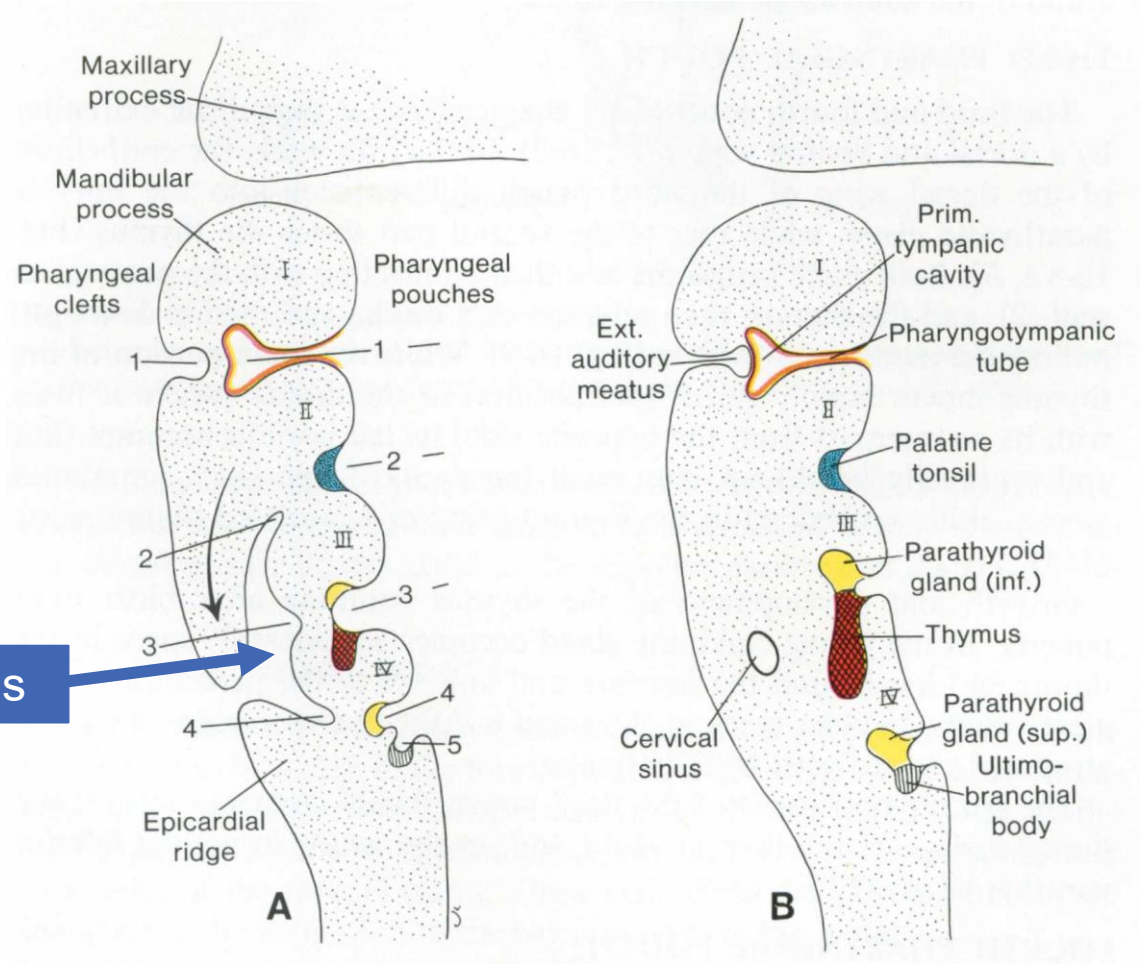
Fig. 17.11 Migration of the thymus, parathyroid glands and the ultimobranchial body
Langman's Medical Embryology,13e.

Branchial Cleft Cyst (Lateral Cervical Cyst)

- Persistent 2nd cleft
- Branchial cleft cyst, sinus, or fistula
- Lateral neck mass, anterior to SCM



Cervical sinus



Cyst = closed cavity

Sinus = one opening (external or internal)

Fistula – two openings (skin ↔ pharynx)

Fig. 17.10 *Langman's Medical Embryology*,13e.

Thyroid Development

- **Origin:** Endodermal evagination (week 4) at foramen cecum (tongue base)
- **Migration:** Descends via thyroglossal duct
- **Final Position:** Duct degenerates → thyroid isolated in neck
- Not derived from pharyngeal arches

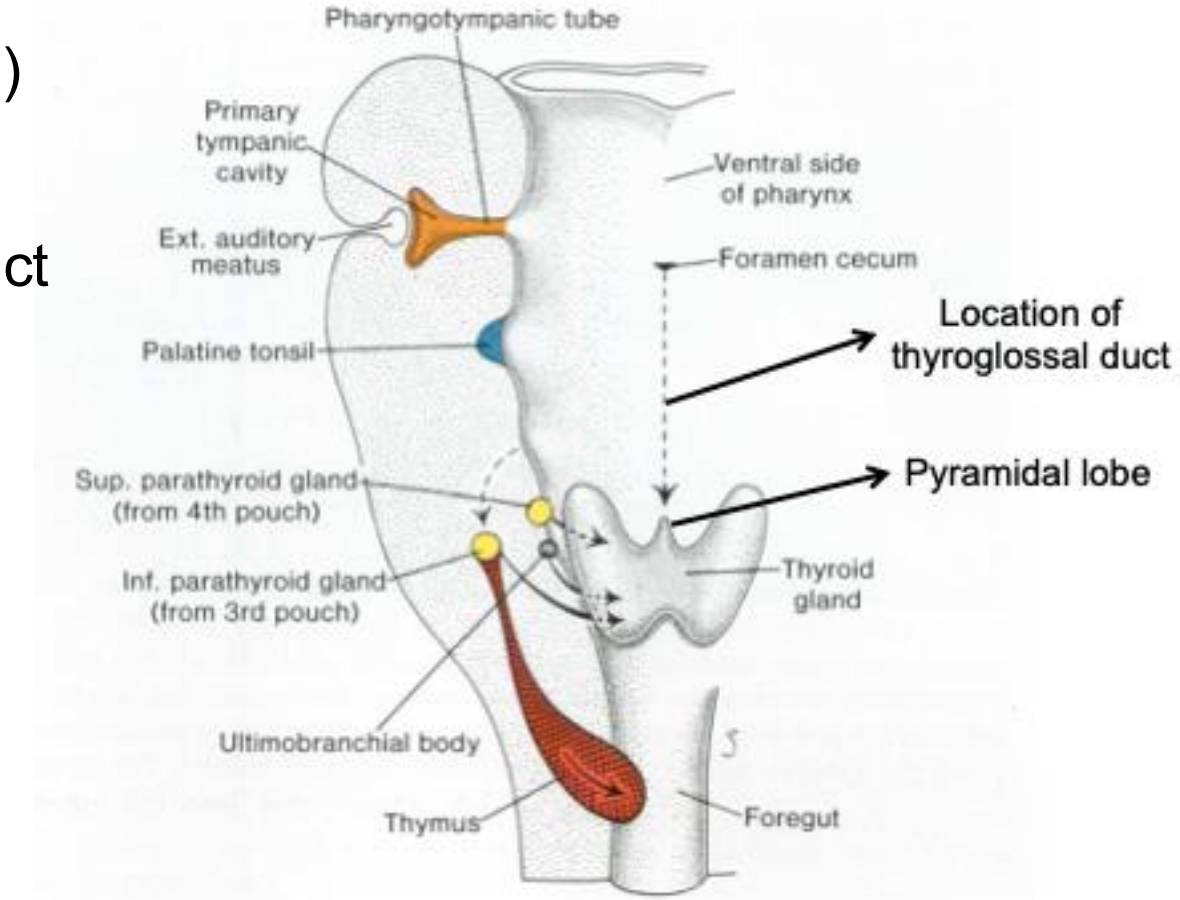


Fig. 17.11 Migration of the thymus, parathyroid glands and the ultimobranchial body *Langman's Medical Embryology*,13e.

Clinical: Thyroid Development

Thyroglossal Duct Cyst

- Failure of thyroglossal duct regression
- Midline mass, moves with swallowing/tongue protrusion

Ectopic/Lingual Thyroid

- Thyroid tissue at tongue base

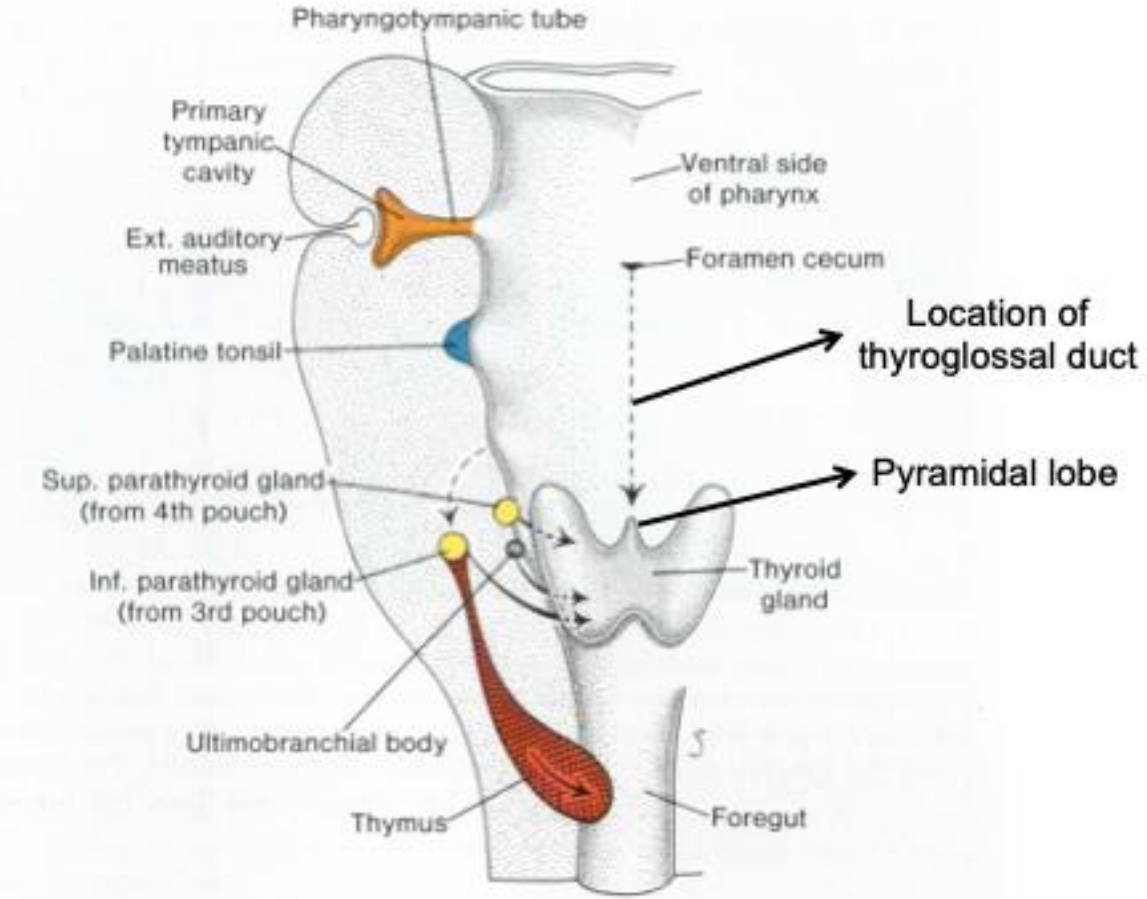
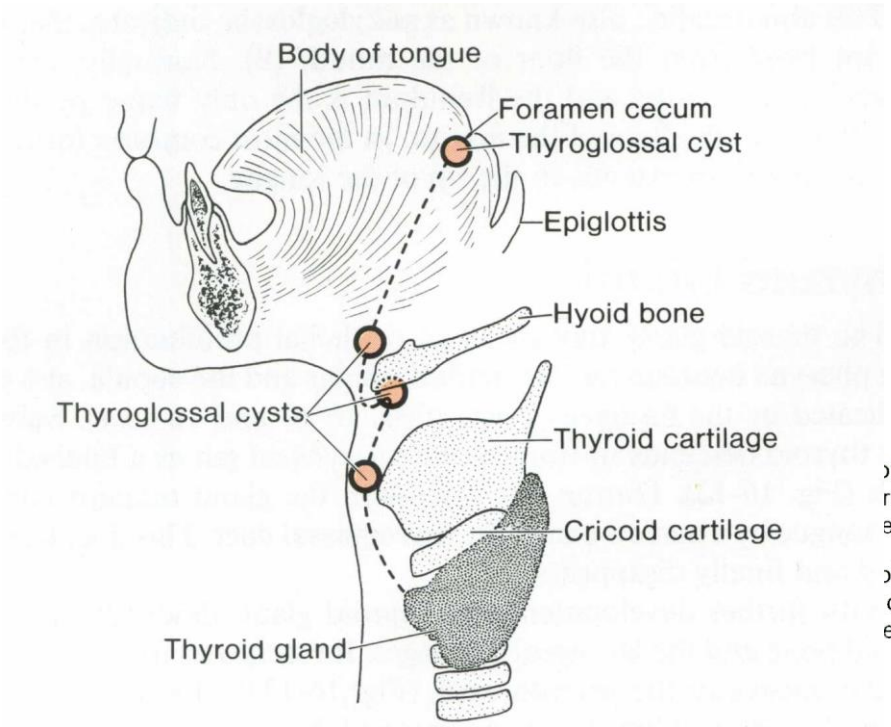
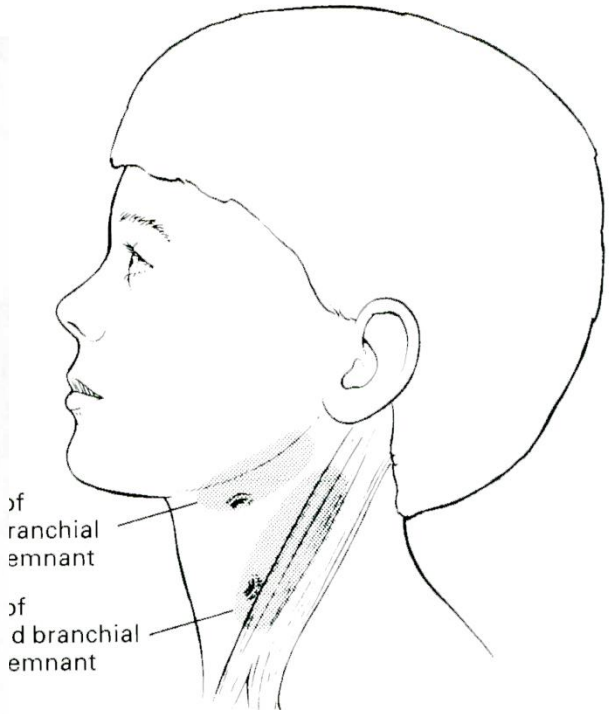


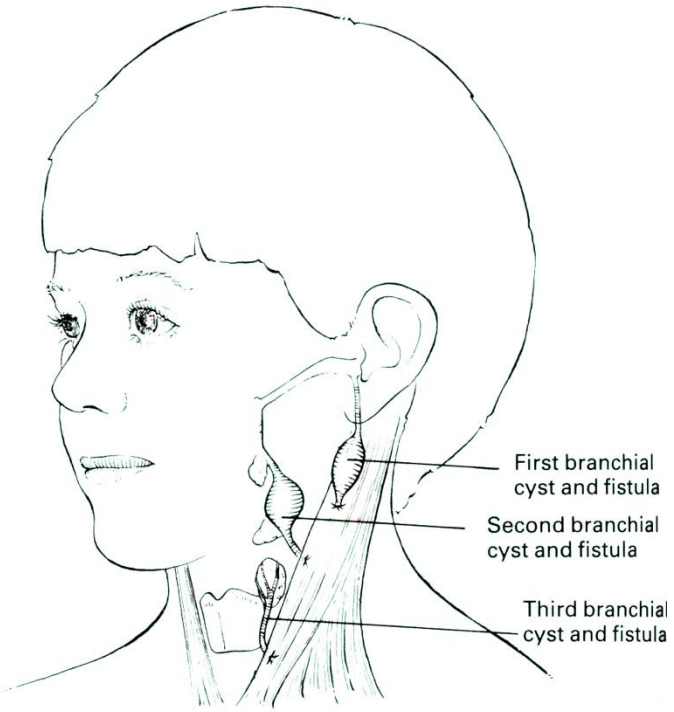
Fig. 17.11 Migration of the thymus, parathyroid glands and the ultimobranchial body *Langman's Medical Embryology*,13e.

Thyroglossal Duct Cyst vs. Branchial Cleft Cyst

- Thyroglossal duct cyst: Midline, moves with swallowing
- Branchial Cyst: Lateral neck, does not move with swallowing

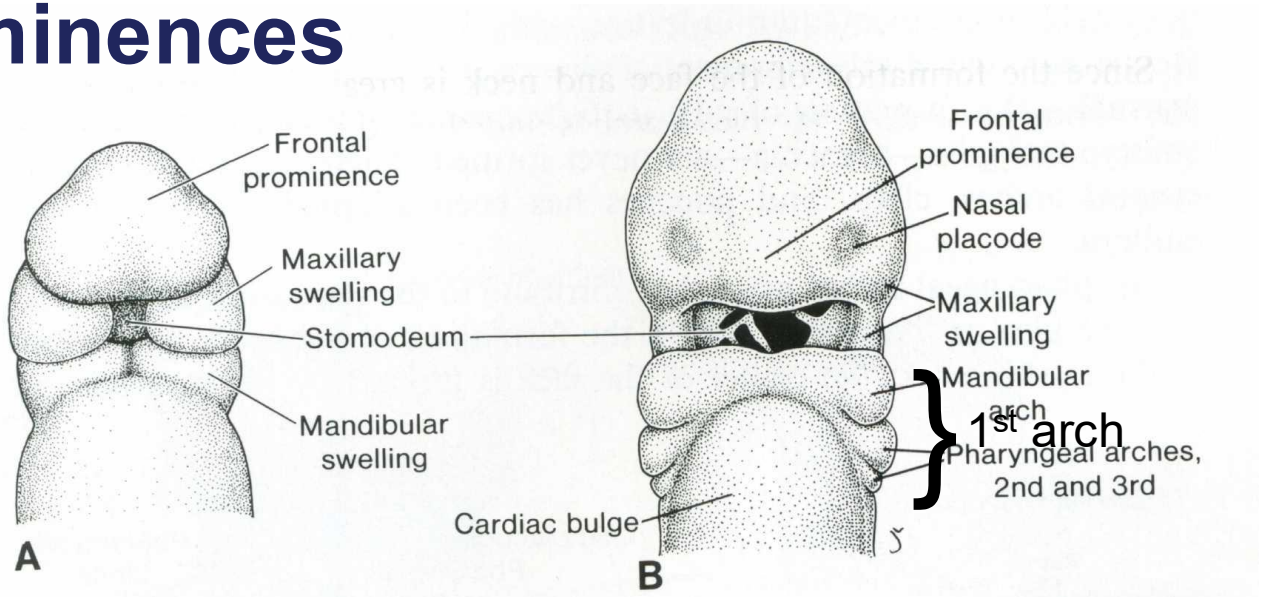






Development of the Face

Facial Prominences

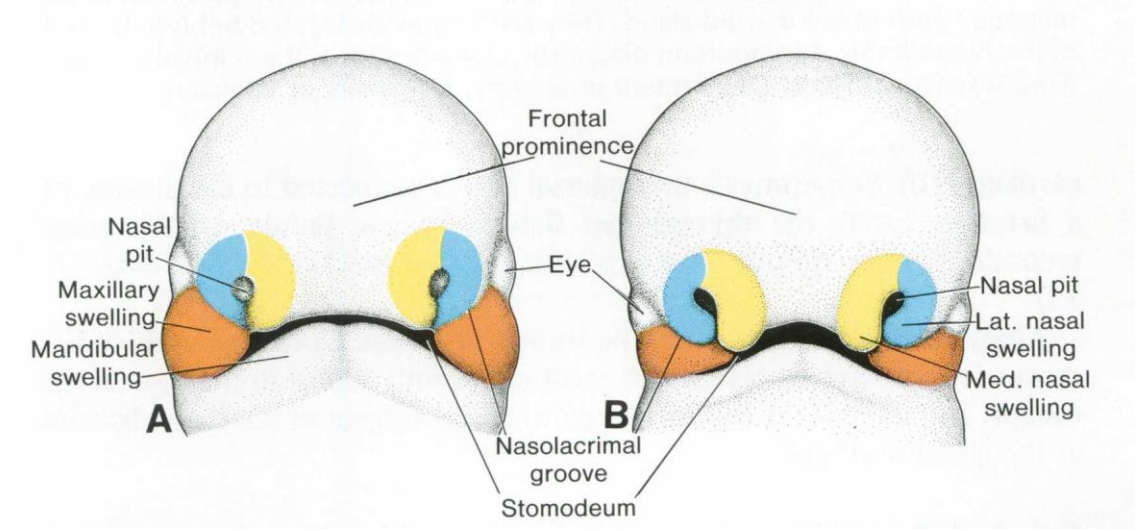


Frontal views of embryo at the end of the 24 Day (A) and 30 days (B)

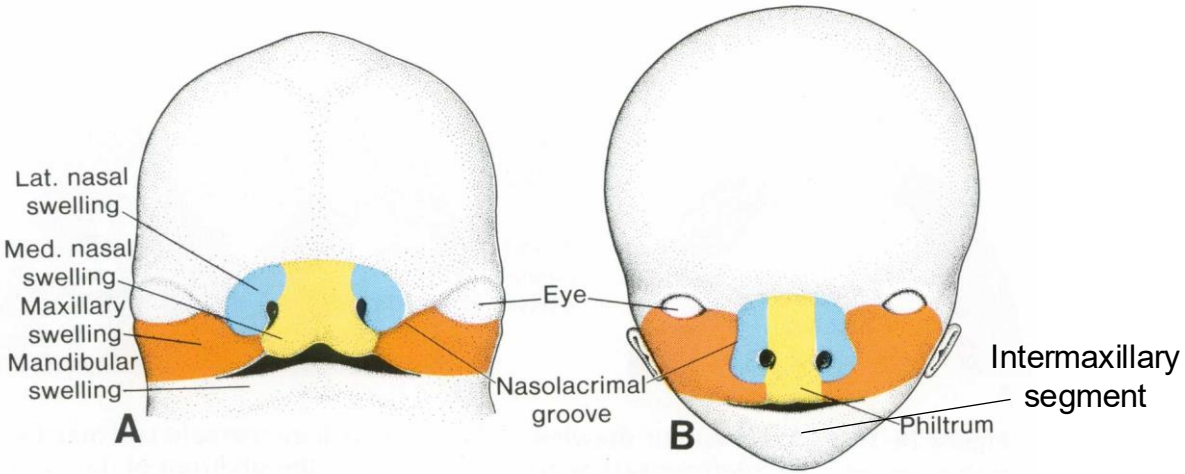
Failure of Fusion
→ cleft lip, cleft palate, facial clefts

- Face forms from 5 prominences
- Frontonasal → forehead, nose, philtrum
- Maxillary (2) → cheeks, lateral portions of upper lip
- Mandibular (2) → lower lip, mandible
- Medial nasal prominences → merge to form the intermaxillary segment, including the philtrum

Development of the Face



Face Development: 5th and 6th week



Face Development: 7th and 10th week

- Nasal placodes develop into medial and lateral nasal prominences
- Medial nasal prominences merge → intermaxillary segment (philtrum)
- Maxillary + medial nasal prominences → formation of upper lip
- Lateral nasal prominences → sides (alae) of nose
- Failure of maxillary–medial nasal fusion → cleft lip

Medial nasal + maxillary prominences → upper lip | Failure of fusion → cleft lip

Fig. 17.22, 17.23. LME

Palate Formation

Primary Palate

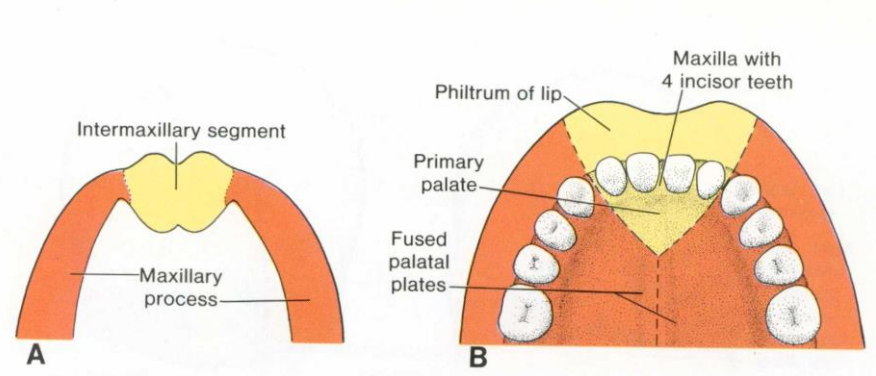
- Develops from the intermaxillary segment (merged medial nasal prominences)
- Forms the small anterior portion of the hard palate, anterior to the incisive foramen

Secondary Palate

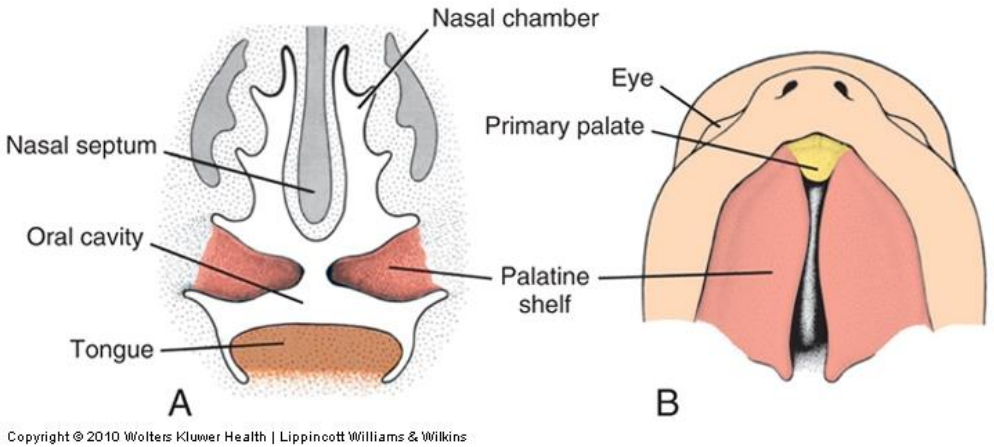
- Develops from palatal shelves of the maxillary prominences
- Shelves elevate and fuse at the midline
- Forms the remaining hard palate and soft palate

Final Fusion

- Secondary palate joins the primary palate and nasal septum



Primary and Secondary Palate



Fusion of Nasal Septum with Palate

Primary palate = medial nasal prominences | Secondary palate = maxillary prominences

Palate Formation: Stepwise Sequence

Step 1 — Initial Position

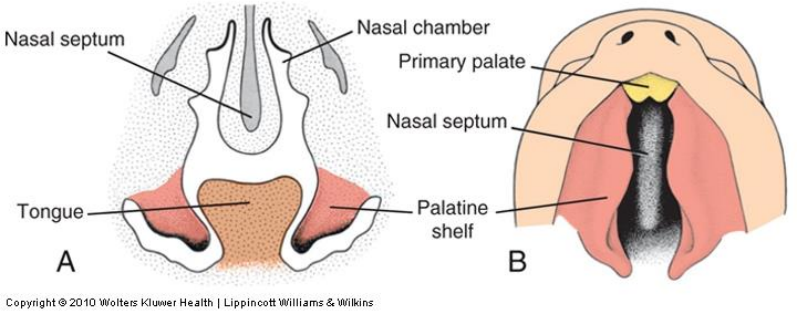
- Palatal shelves grow vertically alongside the tongue

Step 2 — Elevation

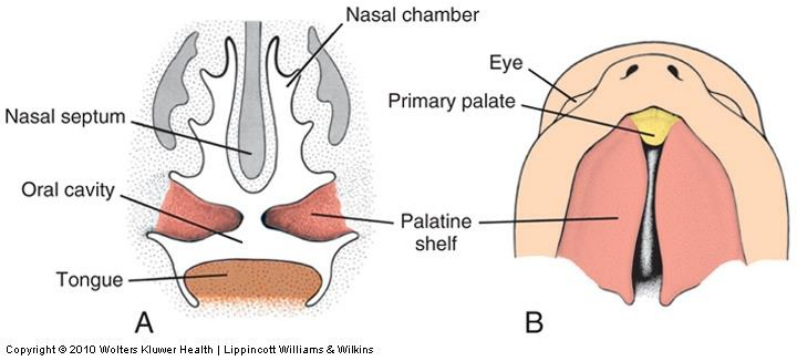
- Tongue descends; palatal shelves elevate into a horizontal position

Step 3 — Fusion

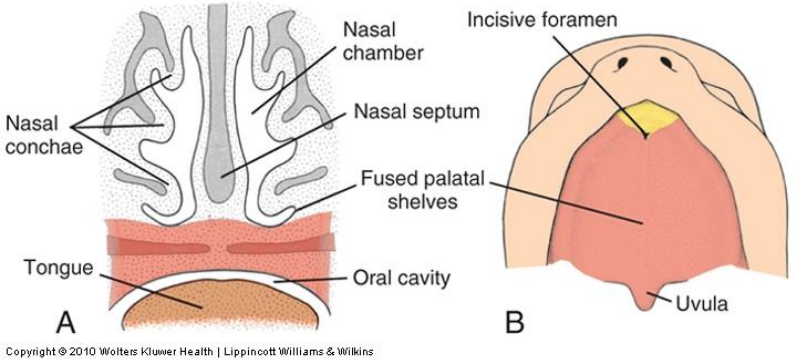
- Shelves fuse at the midline and with the primary palate and nasal septum
- Separates the developing oral and nasal cavities



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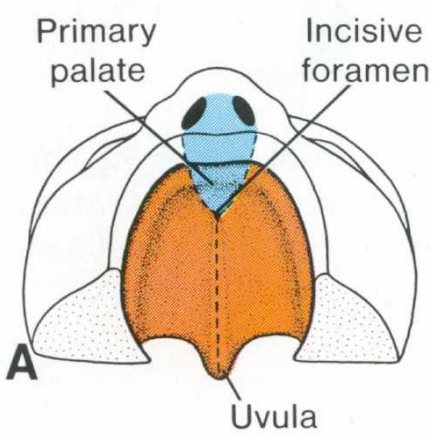
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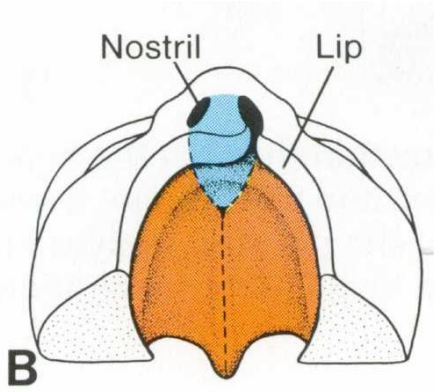
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Normal and Abnormal Development of the Lip and Palate

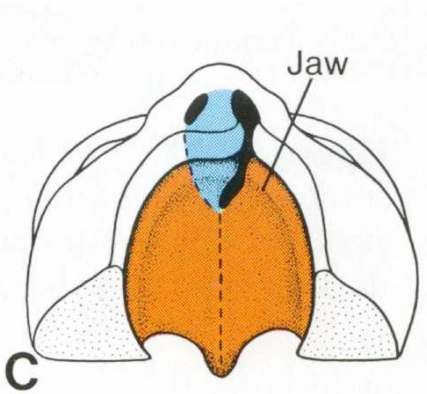
- **Cleft Lip:** failure of fusion between maxillary and medial nasal prominences
- **Cleft Palate:** failure of palatal shelf fusion or fusion with the primary palate/nasal septum
- **Clinical Consequences:** feeding difficulties, nasal regurgitation, and speech abnormalities



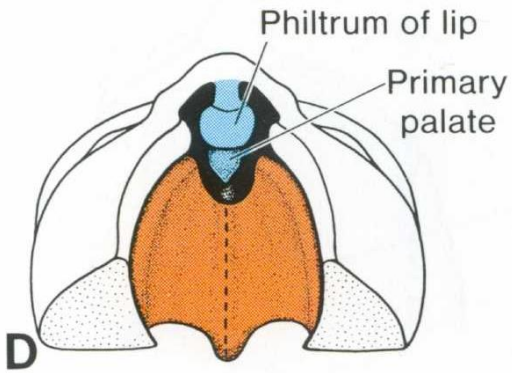
Normal Lip and Palate



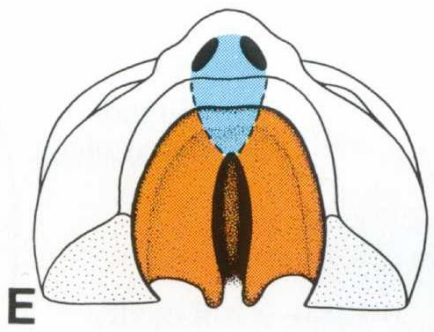
Unilateral CL extending in the nose



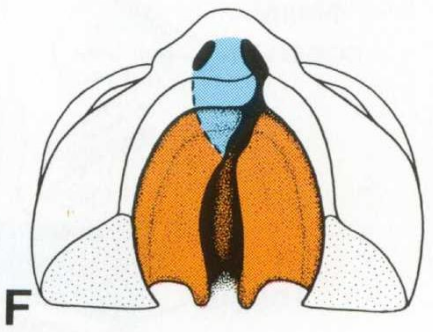
Unilateral CL involving the lip and jaw and extending to the incisive foramen



Bilateral CL involving the lip and jaw



Isolated CP



CP combined with unilateral CL

Cleft lip = facial prominence fusion defect
Cleft palate = palatal fusion defect

LME, Fig. 17.28

Cleft Lip and Palate (Facial Clefts)

Cleft Lip

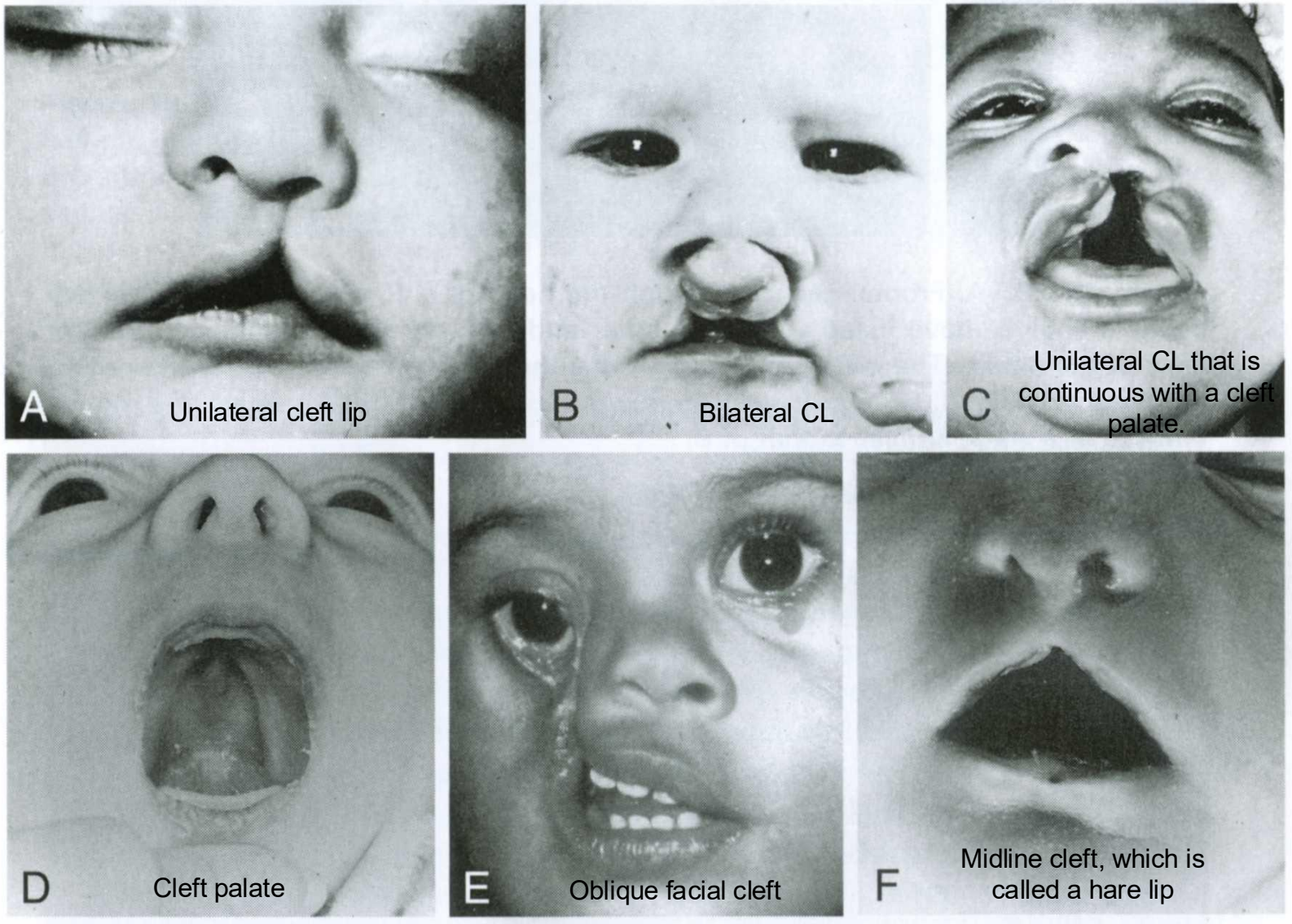
medial nasal +

maxillary fusion failure

Cleft Palate

Palatine shelves

fail to fuse



Developmental Neuroanatomy & Congenital Disorders

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